

CENG381 Stochastic Processes

Wald's Equality — Practice 1

Q1 (30 points). Wald's Equality (Theorem 4.5.1): Let $\{X_n; n \geq 1\}$ be i.i.d. with mean X . If J is a stopping time for $\{X_n\}$ with $E[J] < \infty$, then:

$$E[S_J] = X \cdot E[J], \quad \text{where } S_J = X_1 + \dots + X_J$$

- a) Explain in plain words what a 'stopping time' is. Give one example of a valid stopping time and one example that is NOT a valid stopping time for a sequence $X_1, X_2, \dots$
 - b) Let $X_1, X_2, \dots$ be i.i.d. Uniform $\{1, 2, 3, 4, 5, 6\}$ dice rolls (mean $X = 3.5$). A player stops at $J =$ first roll n such that $S_n \geq 21$. Simulations suggest $E[J] \approx 6$. Use Wald's equality to estimate $E[S_J]$. Does the result make intuitive sense for this game?
 - c) Let $X_i \sim \text{Exp}(\lambda)$ (mean $1/\lambda$). A stopping time $J = N(c) + 1$ counts how many arrivals are needed to first exceed a threshold c (where $N(c)$ is the number of arrivals by time c). For large c , the elementary renewal theorem gives $E[N(c)] \approx \lambda c$. Approximate $E[J]$ and then use Wald's equality to find $E[S_J]$. Interpret the overshoot $E[S_J] - c$.
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Q2 (35 points). A biased coin gives $+1$ w.p. p and -1 w.p. $q = 1 - p$, with $p > 1/2$. Let $S_n = X_1 + \dots + X_n$ ($S_0 = 0$). Define $J = \min\{n \geq 1 : S_n = 1\}$ (first time the gambler is ahead by 1).

- a) Let $\theta = P(J < \infty)$. Derive the equation $\theta = p + q\theta^2$ by conditioning on the first flip: either $S_1 = 1$ (done, w.p. p) or $S_1 = -1$ (w.p. q), after which the gambler must make two independent comebacks from -1 to 0 and from 0 to 1 , each succeeding w.p. θ .
 - b) Solve the quadratic $(1 - p)\theta^2 - \theta + p = 0$. For $p > 1/2$, show that both roots are $\theta_1 = 1$ and $\theta_2 = p/q$. Explain why we must choose $\theta = 1$ when $p > 1/2$ (Hint: for $p > 1/2$, $p/q > 1$ which is not a valid probability). What is θ when $p < 1/2$?
 - c) For $p > 1/2$, $E[J]$ satisfies $E[J] = p \cdot 1 + q \cdot (1 + 2E[J])$. Solve to find $E[J] = 1/(2p - 1)$. Then verify Wald's equality: $E[S_J] = E[X] \cdot E[J] = (2p - 1) \cdot 1/(2p - 1) = 1 \checkmark$. For $p = 2/3$: compute θ , $E[J]$, and $E[S_J]$. What happens to $E[J]$ as $p \rightarrow 1/2$ from above?
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Q3 (35 points). A quality inspector samples items one by one. Each item is defective independently w.p. $q = 0.2$ (good w.p. $p = 0.8$). Let $X_i = 1$ if item i is defective, 0 if good. The inspector uses two different stopping rules.

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- a) Stopping rule A: stop after finding exactly 4 defective items. This is a Negative Binomial stopping rule: $J_A \sim \text{NegBin}(4, 0.2)$. Compute $E[J_A] = 4/q$. Apply Wald's equality to verify $E[S_{J_A}] = 4$ (four defectives found).
- b) The inspector also counts the number of good items inspected: $G = J_A - S_{J_A}$ (total inspected minus defectives). Find $E[G]$. Also, what is $E[G] / E[J_A]$ (the long-run fraction of good items)? Verify it equals $p = 0.8$.
- c) Stopping rule B: stop at $J_B = \min\{n : S_n \geq 3\}$, but now defect rate changes to $q' = 1/4$. Using Wald's equality, compute $E[J_B]$. Suppose additionally $X_i \sim \text{Bernoulli}(1/4)$ but the inspector stops at the first time $S_n \geq 6$ instead. How does $E[J_B]$ scale with the target?